Product Requirements Document (PRD): Test Scheduling Module

1. Executive Summary

Product Name: Test Scheduling Module
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1.1 Purpose

The Test Scheduling Module is a platform aspect to allow administrators to efficiently schedule tests for students with two methods (by category (board/class assignment) and by user ID selectable by individual students). The module contains unique features including confidence assessment, tagging questions, and syncing admin and student portals.

1.2 Business Objectives

- Streamline test scheduling workflows for educational institutions
- Reduce administrative overhead through automated processes
- Improve student engagement through intuitive interfaces
- Enable data-driven insights through confidence assessment and analytics

2. Product Overview

2.1 Key Features

- **Dual Scheduling Methods:** Board/class-based and individual student selection
- **Multiple Upload Options:** CSV, Excel, text files, manual selection, API integration
- **Confidence Assessment:** Real-time facial expression analysis and post-question surveys
- **Question Tagging System:** AI-powered and manual question categorization
- **Synchronized Portals:** Real-time synchronization between admin and student interfaces
- **Comprehensive Communication:** Multi-channel notification system

2.2 Target Users

- **Primary Users:** Educational administrators, teachers
- **Secondary Users:** Students, parents (notification recipients)
- **System Administrators:** Technical support and maintenance staff

3. User Stories and Requirements

3.1 Admin User Stories

Epic 1: Test Scheduling

- As an admin, I want to schedule tests by board and class so I can efficiently assign test to entire cohorts
- As an admin, I want to schedule tests for individual students using their user IDs so I can create targeted assessments
- As an admin, I want to upload student lists via multiple file formats so I can work with existing data systems
- As an admin, I want to configure test parameters so I can customize assessment conditions

Epic 2: Test Management

- As an admin, I want to view all scheduled tests so I can monitor upcoming assessments
- As an admin, I want to edit or cancel scheduled tests so I can respond to changing requirements
- As an admin, I want to monitor real-time participation so I can track student engagement

- As an admin, I want to export test reports so I can analyze performance data

Epic 3: Question Management

- As an admin, I want to review and edit question tags so I can ensure accurate categorization
- As an admin, I want to see AI confidence scores for tags so I can focus on areas needing attention
- As an admin, I want to bulk edit question metadata so I can efficiently manage large question sets

3.2 Student User Stories

Epic 1: Test Access

- As a student, I want to receive notifications about scheduled tests so I can prepare adequately
- As a student, I want to access tests from my dashboard so I can view upcoming assessments
- As a student, I want to view test details so I can understand requirements and expectations

Epic 2: Test Taking

- As a student, I want to take tests within specified time windows so I can demonstrate my knowledge
- As a student, I want my progress to be auto-saved so I don't lose work due to technical issues

Epic 3: Results and Feedback

- As a student, I want to view my test results so I can understand my performance
- As a student, I want to see my confidence scores after the completion of the assessment so I can reflect on my test-taking approach

4. Functional Requirements

4.1 Admin Portal Requirements

4.1.1 Dashboard

- Display quick statistics: active tests, total students, pending reviews, success rates
- Navigation menu with clear sections: Schedule Test, Manage Tests, Calendar, Reports, Settings, Students, Notifications

4.1.2 Test Scheduling Interface

- **Category Selection (Board & Class):**
    - Interactive card interface for board selection (CBSE, ICSE, State Board)
    - Cascading class selection with section options
    - Student filtering by gender, stream, custom tags
    - Preview functionality for selected student cohorts
- **Individual Student Selection:**
    - Manual search and selection interface
    - CSV upload with validation and error handling
    - Excel file upload with template support
    - Text file upload with flexible formatting
    - Copy-paste functionality for user ID lists
    - Group/tag import from existing classifications
    - API integration for external system connectivity
    - QR/barcode scanning capabilities

4.1.3 Test Configuration

- **Test Selection:** Card-based interface for existing tests with interactive design
- **Test Creation:** Form-based interface for new test creation
- **Execution Parameters:**
    - Test window configuration (start/end date and time)
    - Duration settings and attempt limits
    - Access controls (password, IP restriction, device limitation)
    - Randomization options for questions and answers
    - Special accommodations for accessibility

4.1.4 Question Tagging System

- **Metadata Fields:**
    - Topic/Subject Area
    - Subtopic classification
    - Difficulty Level (Easy, Medium, Hard)
    - Question Type (MCQ, Descriptive, Others)
    - Cognitive Level (Recall, Application, Analysis, Evaluation)
    - Marks/Weightage
    - Estimated Time to Solve
    - Bloom's Taxonomy Level
    - Language/Medium
    - Tag Confidence Score
- **Interface Features:**
    - Table view with all questions and metadata
    - MCQ options display (A, B, C, D) with correct answer indication
    - Inline editing capabilities
    - Bulk editing for multiple questions
    - Filtering and search functionality
    - AI confidence indicators
    - Progress tracking for review completion

4.2 Student Portal Requirements

4.2.1 Dashboard

- Upcoming tests display with key information
- Active tests with start buttons
- Completed tests with results links
- Notification center for test-related communications

4.2.2 Test Interface

- **Pre-Test:**
    - Test details display (subject, duration, instructions)
    - Resource downloads (attachments, reference materials)
    - Reminder setting options
    - Calendar integration
- **During Test:**
    - Question navigation interface

        - Progress indicator
        - Auto-save functionality
        - Timer display
        - Confidence assessment integration
    - **Post-Test:**
        - Submission confirmation
        - Results display (when available)
        - Confidence score presentation
        - Feedback access

4.3 Confidence Assessment System

4.3.1 Data Collection

- **Facial Expression Analysis:**
    - Real-time webcam capture (with consent)
    - AI-powered emotion detection
    - Confidence score calculation based on facial cues
- **Survey Integration:**
    - Post-question confidence surveys
    - Customizable survey questions
    - Response aggregation and analysis

4.3.2 Score Calculation

- Combined facial analysis and survey data
- Per-question confidence scoring
- Overall test confidence aggregation
- Confidence trend analysis

4.3.3 Display Integration

- Student result pages with confidence scores
- Admin analytics with confidence metrics
- Question-level confidence indicators
- Dashboard confidence summaries

5. Technical Requirements

5.1 Architecture

- **Frontend:** Modern web application with responsive design
- **Backend:** RESTful API with microservices architecture
- **Database:** Relational database with real-time synchronization
- **File Storage:** Cloud-based storage for uploads and resources
- **AI/ML:** Confidence assessment and question tagging models

5.2 Performance Requirements

- **Response Time:** < 3 seconds for all major operations
- **Concurrent Users:** Support for 1000+ simultaneous users
- **Data Processing:** Handle 500+ questions in tagging interface
- **File Upload:** Support files up to 10MB with progress indicators

- **Real-time Sync:** < 2 seconds for admin-student portal synchronization

5.3 Security Requirements

- **Authentication:** Multi-factor authentication for admin users
- **Authorization:** Role-based access control (RBAC)
- **Data Protection:** End-to-end encryption for sensitive data
- **Privacy Compliance:** GDPR and FERPA compliance for student data
- **Audit Trail:** Comprehensive logging for all user actions

5.4 Integration Requirements

- **Student Information Systems (SIS):** API integration for student data
- **Learning Management Systems (LMS):** Seamless data exchange
- **Notification Services:** Email, SMS, push notification providers
- **Calendar Systems:** Integration with Google Calendar, Outlook
- **External Authentication:** SSO integration with institutional systems

6. User Interface Requirements

6.1 Design Principles

- **Accessibility:** WCAG 2.1 AA compliance
- **Responsive Design:** Mobile-first approach with desktop optimization
- **Usability:** Intuitive navigation with minimal learning curve
- **Consistency:** Unified design language across all interfaces
- **Performance:** Optimized for fast loading and smooth interactions

6.2 Visual Design

- **Color Scheme:** High contrast with accessibility considerations
- **Typography:** Clear, readable fonts with appropriate sizing
- **Icons:** Consistent iconography with meaningful visual cues
- **Layout:** Grid-based design with proper spacing and alignment
- **Interactive Elements:** Clear hover states and feedback mechanisms

6.3 User Experience

- **Navigation:** Logical flow with breadcrumbs and clear pathways
- **Feedback:** Immediate response to user actions with appropriate messaging
- **Error Handling:** Clear error messages with actionable guidance
- **Help System:** Contextual help and comprehensive documentation
- **Personalization:** Customizable dashboards and preference settings

7. Non-Functional Requirements

7.1 Reliability

- **Uptime:** 99.9% system availability
- **Data Integrity:** Zero data loss with automated backups
- **Fault Tolerance:** Graceful degradation during system issues
- **Recovery:** < 4 hours recovery time for major incidents

7.2 Scalability

- **User Growth:** Support for 10x user base expansion

- **Data Volume:** Handle increasing question and result datasets
- **Geographic Distribution:** Multi-region deployment capability
- **Feature Expansion:** Modular architecture for future enhancements

7.3 Maintainability

- **Code Quality:** Well-documented, modular codebase
- **Version Control:** Git-based workflow with proper branching
- **Testing:** Comprehensive unit, integration, and end-to-end testing
- **Deployment:** Automated CI/CD pipeline with staging environments

8. Success Metrics

8.1 User Adoption

8.2 System Performance

9. Implementation Timeline

10. Risk Assessment

10.1 Technical Risks

- **AI Model Accuracy:** Confidence assessment may require extensive training data
- **Integration Complexity:** External system connections may face compatibility issues
- **Performance Scalability:** Large file uploads may impact system performance
- **Data Privacy:** Handling sensitive student data requires careful compliance management

10.2 Mitigation Strategies

- **Iterative Development:** Regular testing and feedback incorporation
- **Fallback Systems:** Manual overrides for AI-powered features
- **Performance Monitoring:** Real-time system monitoring and alerting
- **Security Audits:** Regular security assessments and vulnerability testing

- Security protocols

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